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RESEARCH

Navigating social debt and its link with technical debt in large-scale agile software development projects

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Abstract
Agile methodologies have emerged as transformative paradigms in the ever-evolving software development landscape, emphasizing iterative development, customer collaboration, and adaptability. As the scope and complexity of projects and organizations expand, applying agile principles within the context of Large-Scale Agile Development (LSAD) encounters distinctive challenges. The majority of challenges encountered in LSAD, technical and non-technical, are attributed to the accrual of social debt. However, a conspicuous gap remains in understanding and addressing social debt in LSAD. This study aims to fill this void by investigating social debt in LSAD through an in-depth industrial case study with a leading Nordic company specializing in telecommunications software and services and focusing on producing secure 5G network solutions. The study investigates the causes of LSAD's social debt and examines its impacts on secure 5G telecom software development. By addressing these objectives, this research sheds light on a critical aspect of LSAD's social debt, caused by 3C challenges(communication, coordination and collaboration), social confines challenges, community smells challenges, and organisational social challenges in the telecom sector that have been underrepresented in the existing literature.

Keywords Agile · Large-scale scale development · Scrum · Technical debt · Social debt · Non-technical debt · Telecom sector · Secure solutions · 5G · Large-scale agile development challenges · Software project management · Business support

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1 Introduction

Agile methods were designed for small-scale projects with self-organising teams. However, the potential benefits of agile methodologies have made them appealing to large projects and companies Dikert et al. (2016) Kowalczyk et al. (2022). Because agile methods tend to scale (can scale with certain limitations), many companies are switching to an agile way of working on large projects Edison et al. (2021) Das and Gary (2021). Agile project success relies on social factors, including end-user participation, face-to-face collaboration, regular feedback, and transparency. However, applying these social aspects to larger teams, projects, and organizations is difficult Dikert et al. (2016). The negative consequences of weakened social factors in software development are called social debt (SD) Tamburri (2019). Social debt is defined as a "cumulative and increasing cost in the current state of things, connected to invisible and negative effects within a development community" Martini et al. (2019). In LSAD, big teams and companies suffer from social debtMartini et al. (2019). This makes large-scale agile implementation difficult. Bad communication generates social debt, delays, rework, and decreased production. Ineffective team or departmental collaboration can result in redundancy, competing priorities, and inefficiency. Cultural differences between teams or departments might inhibit agile deployment and produce social debt Martini et al. (2019). To succeed, companies must manage social debt, which limits large-scale agile initiatives' agility and capacity Martini et al. (2019). Irrespective of all the mentioned LSAD challenges, practitioners are adopting agile methods for large projects to follow the fast-paced nature of the software industry Das and Gary (2021). So, for the better adoption of LSAD and to obtain all its fruitful outcomes, it is essential to analyse social debt in the context of LSAD. But even though sociotechnical factors and debts play a huge role in software development, new research has only revealed about the technical debt (TD) phenomenon. TD refers to less-than-ideal technical implementations that help in the short term but create a situation where a long-term interest is paid Besker et al. (2019); Li et al. (2015). However, there is a lack of research and information about other types of debt that are socio-technical, especially SD Saeeda et al. (2023); Ahmad and Gustavsson (2022); Martini et al. (2019).

LSAD is inherently socio-technical. It integrates human collaboration, decisions, planning, and management with software development processes, coding, and designing. Socially, LSAD emphasizes teamwork, stakeholder engagement, and adaptive change. It focuses on interactions and communication among diverse team members Saeeda et al. (2023). Technically, it incorporates practices like continuous integration and automated testing to handle software complexity. In large-scale settings, this approach requires balancing complex team coordination with technical excellence and aligning agile practices with overarching business strategies to manage multiple teams and maintain agility Tamburri et al. (2013). However, most researchers only research technical aspects such as code quality, design issues, or testing challenges, and social debt analysis is still a topic of much unexplored research Rios et al. (2018). For instance, software engineering practitioners still lack a way to formalize the causes and effects of socio-technical factors and measure the debt (if any) connected to them in software development Tamburri et al. (2013). TD can often be quantified in terms of a monetary value Saeeda et al. (2023). This quantification is typically based on factors like the cost of additional resources required to fix the suboptimal code, the extra effort needed for future modifications, or the potential loss of revenue due to decreased efficiency or system outages Tamburri et al. (2013). Estimation tools and metrics, such as code complexity analyses, can be used to assign a tangible value to technical debt Saeeda et al. (2023). Unlike technical debt, social debt is not easily quantifiable in monetary terms Tamburri et al. (2013);

Saeeda et al. (2023). It's measured in terms of its effects on the team and the project, such as decreased team morale, reduced productivity, increased turnover, or poor product quality. These impacts are often qualitative and can vary widely between teams and organizations Saeeda et al. (2023); Tamburri and Di Nitto (2015). Social debt requires a more nuanced approach to identification and management, often involving surveys, interviews, and observation to understand and address its root causes and effects Saeeda et al. (2023); Tamburri et al. (2013); Kazman (2019). Therefore, there needs to be more investigation on the causes and effects of social debt in the LSAD context for better mitigation and management Saeeda et al. (2023); Ahmad and Gustavsson (2022); Martini et al. (2019). There is a literature scarcity on social debt analysis in LSAD, and the current state-of-the-art literature Saeeda et al. (2023); Ahmad and Gustavsson (2022); Martini et al. (2019); Tamburri et al. (2015) calls for more empirical studies on the topic.

A recent study Martini et al. (2019) shows that agile retrospectives within software teams address more than TD, including nontechnical issues impacting team performance. The latest research Saeeda et al. (2023); Ahmad and Gustavsson (2022) shows it is critical to assess and manage different types of non-technical debts (i.e., SD), as evidence shows that this kind of debt is directly linked to the generation of TD, too Tamburri et al. (2013). So, ignoring any one side will bring equally damaging consequences for the software projects, as caused by TD. Despite this evidence on SD severity, there is a knowledge gap about SD, how it is caused, its effects, how it is linked to TD, and what possible mitigation strategies we can adopt to manage TD in LSAD Martini et al. (2019); Saeeda et al. (2023); Ahmad and Gustavsson (2022).

Our case study aims to cover the knowledge gap by providing industrial empirical evidence on the topic. To achieve our study goals, we are investigating the following research questions:

1. **RQ:** *What are the types of SD, their causes, and consequences in LSAD?*
2. **RQ:** *How does SD lead to the accumulation of TD in LSAD?*
3. **RQ:** *How can SD be mitigated in LSAD?*

This case study aims to answer our research questions using semi-structured interviews. It investigates the types of SD in LSAD. The study provides a detailed analysis of the identified SD causes and their effects on LSAD. It examines the link between various SD types and TD. The study also provides a detailed discussion of how these identified SD types and generated TD affect the production of secure 5G telecom software development in Case X. Further, we are providing mitigation strategies for identified SD challenges from the existing state-of-the-art literature.

The study is organized into ten sections: Introduction (1), Background (2), Research Methodology (3), Results (4), Validity and Limitations (5), Discussion (6), Research Implications (7), Practice Implications (8), Conclusion (9), and Future Work (10).

2 Background

There are several studies conducted by Tamburri (2019); Tamburri and Di Nitto (2015); Tamburri et al. (2015) investigating social aspects of debt in software development under the term "social debt". Social debt decisions are about people and their interactions Tamburri et al. (2015), as decisions related to social debt revolve around people and their interactions. It is also a crucial consideration in LSAD, where effective

teamwork and collaboration are essential Martini et al. (2019). Social debt significantly impacts both the individuals involved in software development tasks and the overall quality of the software they deliver Tamburri et al. (2015), Dybå and Dingsøyr (2008).

Common reasons for social debt accumulation in software development, as highlighted in recent literature Ahmad and Gustavsson (2022); Saeeda et al. (2023), include gender inequalities, communication, collaboration and coordination challenges, community smells, organizational issues, and social confines. Effective communication, collaboration, and coordination, known as the "3C challenges," are crucial for successful social interactions and task completion. Shortcomings in communication, compromises, and negative behaviours can lead to social debt in human interactions. Remote communication presents challenges due to the absence of physical cues, potentially resulting in communication gaps. Insufficient informal communication, like unstructured conversations, can also contribute to social debt.

Caballero-Espinosa et al. (2023) define social debt (SD) within software development teams as socio-technical anti-patterns that disrupt team interactions and behaviours, harming productivity and well-being. Similar to "code smells," SD highlights problematic areas, but instead of technical aspects, it focuses on social and organizational aspects. Examples include lack of trust, communication gaps, and unclear roles. SD undermines teamwork dynamics; for instance, a lack of trust hinders collaboration and communication, leading to decreased productivity and morale. Unclear roles cause confusion and duplicate efforts. SD leads to suboptimal project outcomes, with communication gaps causing misunderstandings and delays. Overlooking or inadequately addressing team-related issues compounds the negative impact of SD over time.

Tamburri et al. (2013) assert that SD and TD stem from decisions made during software development, but SD is distinguished by its dynamic and unpredictable nature due to human factors, complicating its impact assessment. Unlike TD, which is confined to the system codebase and artefacts, SD extends beyond technical decisions, intertwining with team interactions and organizational structures. This broader and more complex nature makes SD a significant factor in software project success Tamburri et al. (2013).

Tamburri (2019) identified social debt (SD) as the additional project cost stemming from suboptimal sociotechnical decisions, such as changes in organizational structure or development methodologies. They emphasized challenges in architecture decision communicability within software development networks, leading to issues like architecture erosion and mistrust. To address this, they introduced the Debt-Aimed Architecture-Level Incommunicability Analysis (DAHLIA) framework, focusing on measuring and quantifying social debt, particularly incommunicability, within a software architecture. Utilizing social network analysis, DAHLIA offers an innovative approach to comprehending and managing social debt intricacies.

Palomba et al. (2017) stress the critical impact of social debt (SD) on software project success, leading to unexpected costs associated with team dynamics. They identify a notable lack of empirical research on SD and its effects on software maintenance and evolution, underscoring the need to investigate the relationship between social and technical debt in software development. The authors propose that analyzing SD, particularly through social network analysis, can enhance software evolution tools, facilitating tasks like code review and maintenance and providing deeper insights into socio-technical aspects of software development.

Martini and Bosch (2017) introduced the CAFFEA framework as a comprehensive solution to address social debt (SD) in software engineering, often stemming from organizational inefficiencies like inadequate collaboration among key roles, leading to architectural technical debt. The framework aims to define architectural responsibilities clearly across

roles and teams, fostering improved collaboration and reducing SD likelihood. By analyzing organizational structure thoroughly, CAFFEA identifies SD areas and those at risk for architectural technical debt, enabling proactive identification and mitigation of social debt, thus preventing costly repercussions.

3 Methodology

A single case study method, as suggested by Runeson and Höst (2009), is used to look into the different types of SD, what causes them, and how they affect LSAD in Case X. Purposeful sampling Maxwell et al. (2008) is used to select experienced software development teams well-versed in scaling agile methods, as detailed in Table 1. This approach is suitable for in-depth analysis of real-world scenarios, especially for addressing specific queries. It allows for a thorough examination of the unique characteristics of Case X, which is essential for understanding diverse LSAD environments. The method excels at collecting qualitative data like interviews, which is crucial for comprehending socio-technical issues such as SD in Case X. This strategy aids in forming hypotheses for future research. It provides actionable insights and recommendations for the studied context. The complete process of the case study followed is depicted in Fig. 1 .

3.1 Case X context

The study focuses on Case X, a leading IT and software service provider in the Nordic region, delivering a wide range of digital services and solutions to both the public and private sectors. Their expertise covers cloud services, cybersecurity, software development (i.e., secure 5G telecom solutions), IT consulting, and advanced technologies like AI and blockchain to support digital transformation. Committed to sustainability, they provide eco-friendly IT solutions while maintaining a global competitive presence and fostering

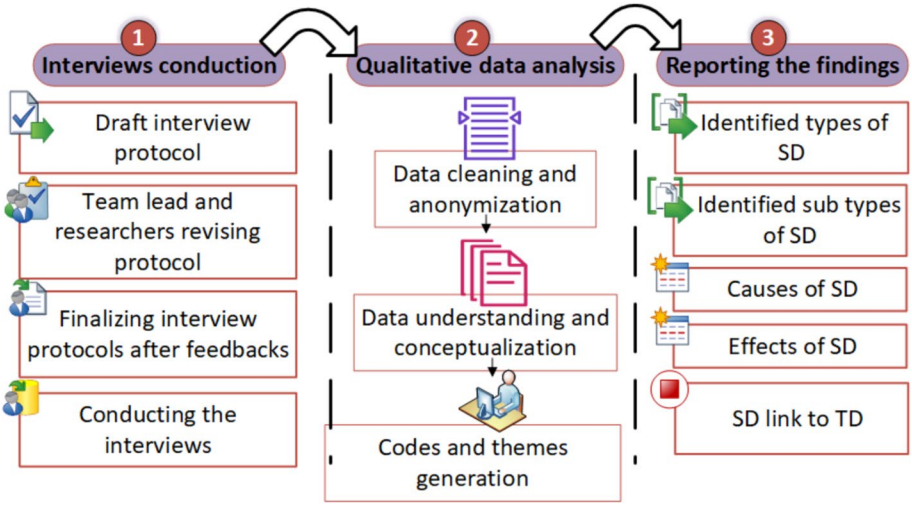

Fig. 1 Research methodology followed

Table 1 SD type (3C challenges): Causes and effects

3C Challenges			
No	**Causes of SD**	**Effects of SD**	**ID**
1	Hierarchical Communication Challenges	Hierarchical communication challenges contradict Agile values, leading to reduced participation, low morale, and cultural discomfort.	P11
2	Remote Inclusive Communication Challenges	Remote Inclusive Communication Challenges result in disengagement, weaker team bonds, reduced collaboration, increased stress, and a downturn in innovation.	P5
3	Lack of Personal Interaction in Inter-team Communication	Lack of Personal Interaction in Inter-team Communication situation hampers the flow of information, coordination, and team alignment.	P2
4	Hard to Communicate People's Needs	Hard to Communicate People's Needs leads to unmet user needs, extra efforts, escalating project costs, scope creep, and stakeholder dissatisfaction.	P1
5	Missing Cross-Program Coordination	Missing Cross-Program Coordination's primary consequence is the duplication of efforts, inefficiencies, missed opportunities, and fosters conflicts.	P7
6	Lack of Coordination for Shared Vision and Participation	Lack of Coordination for Shared Vision and Active Participation in LSAD leads to vision misalignment, eroded trust and morale, and uneven engagement in meetings.	P11
7	Balancing Collaboration in Remote and Hybrid Environments	Balancing Collaboration in Remote and Hybrid Work Environments leads to reduced productivity, hindered innovation, decreased team morale, and compromised trust.	P2

partnerships for innovation. Their primary objectives include reducing operational costs, improving network performance, and expediting solution delivery for network operators. Case X's software business focuses on developing Operations and Maintenance (O &M) solutions for telecom networks, applying infrastructure and network management solutions to their network/NFV testbed, implementing end-to-end (E2E) service management, actively monitoring networks with virtual probes and big data analytics, and collaborating with industry organizations like the European Telecommunications Standards Institute (ETSI) and the Metro Ethernet Forum (MEF) to advance NFV (Network Function Virtualization) for carrier Ethernet 2.0 services. The specific development project under study is a complex, large-scale telecom software system with numerous modules and dependencies. To address the unique needs of its clients, Case X has progressively adopted LSAD. There are twelve software development teams. The teams, typically consisting of six to eight members with diverse roles such as developers, Scrum Masters, Product Owners, designers, and testers, engage in agile ceremonies, including sprint planning, daily standups, retrospectives, and backlog refinement. The teams' workflow emphasizes continuous improvement, technical debt management, and meticulous code reviews, with a focus on self-driven learning and adaptation. The team structures reflect a blend of seasoned and new members, promoting knowledge sharing and collaborative problem-solving, albeit facing challenges in communication and aligning on common goals. The company's approach to agile seems to evolve, with a strong emphasis on practical application and continuous adaptation to suit project and team needs.

3.2 Data collection

For our case study, we conducted 15 semi-structured online interviews via Zoom with software development teams experienced in agile methods. Purposeful sampling was used to select participants from diverse roles, as shown in Table 2. Interviews lasted 60-90 minutes and followed Patton's pragmatic interview guidelines (2014). Each interview was recorded

Table 2 Interviewees background

ID	Interviewee's title	Development Experience
P1	Product owner	10 years
P2	Security master	3 years
P3	Developer	1 year
P4	Developer	10 years
P5	Scrum Master	20 years
P6	Developer	25 years
P7	Project Manager	20 years
P8	Product Guardian	3 years
P9	Scrum Master	1 year
P10	Architect + Scrum Master	16 years
P11	Developer + Scrum master	1 year
P12	Senior software engineer	7 years
P13	Developer	10 years
P14	Developer	12 years
P15	Tester/QA	2 years

and involved one interviewer and one observer. Questions covered various aspects, including the study's purpose, participants' backgrounds, product development, large-scale agile development, experienced social challenges, causes and effects of these challenges, and their relation to technical debt in Case X. Two rounds of data collection occurred, with the first round conducted between September and November 2022. After cleaning and understanding the data, transcripts were sent back to participants for clarification in March 2023. Cleaned transcripts were received in June 2023, with one researcher checking for transcription errors and organizing the data question by question.

3.3 Data analysis

Our study employed the thematic analysis methodology as outlined by Cruzes and Dybå, meticulously following their six-phase process for data analysis. This approach, detailed in their work (2011) and illustrated in Fig. 2, provided a structured and comprehensive framework for examining and interpreting the data gathered in our research. Here's a detailed step-by-step breakdown of how we applied each phase:

1. **Familiarization with the data:** In the first stage of our thematic analysis, we diligently transcribed and anonymized the data, safeguarding participants' confidentiality. This

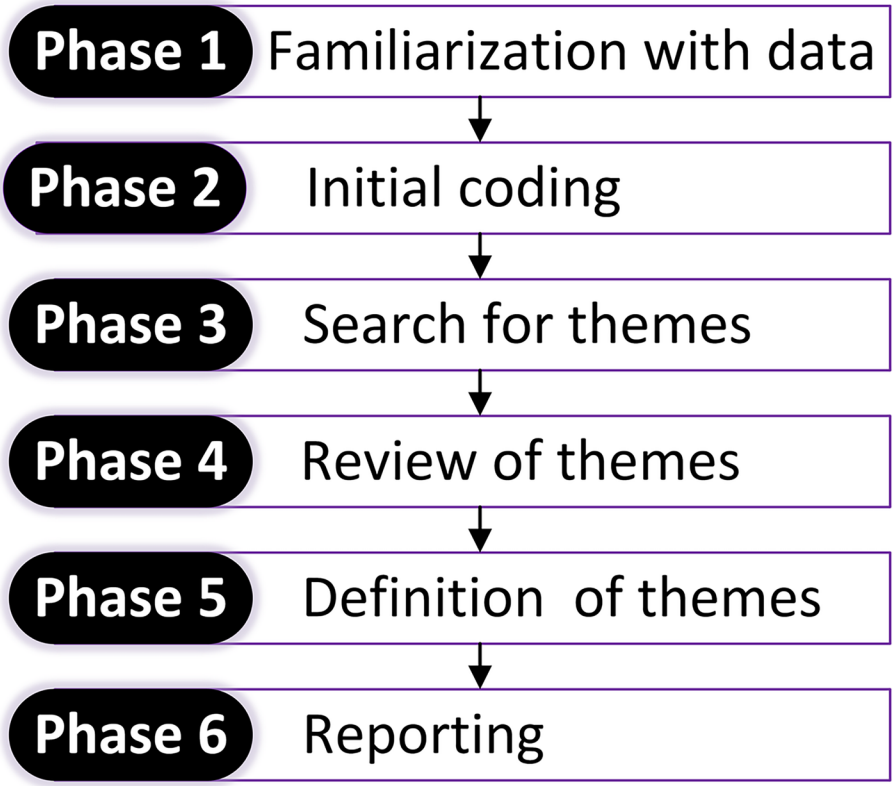

Fig. 2 Thematic analysis steps followed

was vital for upholding ethical standards. We then carefully reviewed each transcript to validate its accuracy, ensuring our analysis was based on precise representations of participants' contributions. Next, we engaged in the familiarization process, immersing ourselves in the data through repeated readings. Our aim was to gain a deep understanding of the content, going beyond surface observations to identify patterns, nuances, and potential themes related to SD in the context of LSAD and its implications.

2. **Coding the data:** During coding, we analyzed the data closely, assigning descriptive labels to relevant sections, resulting in 24 important codes. These codes represent socio-technical challenges in software development teams, such as hierarchical communication, remote communication issues, and a lack of personal interaction. Difficulties in expressing needs, coordination problems, and challenges in remote and hybrid work environments were also identified. Additionally, a pervasive lack of psychological safety was noted among the teams.We observed specific effects like the black cloud effect, organizational silo effect, lone wolf effect, and radio silence effect, along with broader organizational issues such as power distance, fragmentation, and cultural differences. Organizational turbulence caused delivery delays and trade-offs in workflow. Management inefficiencies and the challenge of bridging generational gaps were also noted, as well as the need to adapt to varied role expectations. The social dynamics of task complexity and social constraints were identified, along with the challenge of integrating isolated team members. Each aspect reflects how socio-technical factors impact software development, leading to SD.
3. **Generating initial themes:** In the phase of generating initial themes, we analyzed the 24 assigned codes to uncover broader patterns and relationships, focusing on socio-technical challenges within LSAD leading to SD. We delved into the root causes of SD and explored its ripple effects on LSAD processes, team interactions, and project outcomes. Our analysis revealed themes related to communication, collaboration, coordination challenges, organizational issues, socio-technical decisions, and development team issues. These themes provided a framework for synthesizing our findings in subsequent stages of analysis.
4. **Reviewing and developing themes:** In this stage, we critically evaluate the preliminary themes that emerged from the coding phase. This involved revisiting the data set and the codes to ensure that the themes accurately represented the patterns observed in the data. It was a phase of refinement and validation where some themes were combined(i.e. communication, collaboration and coordination challenges), and some were separated (i.e. team issues or community challenges from individual challenges) or further clarified(to look for more accurate and matching descriptions from interviews) to ensure they captured the essence of the data. During this review, we compare themes against each other and back against the coded extracts to check for coherence and consistency. This iterative process of reviewing led to the reworking of some themes, the discarding of others that lacked sufficient evidence, and the development of new themes that better fit the data.
5. **Refining, defining, and naming themes:** After refining the themes, we assign them concise names reflecting their essence. The thematic analysis resulted in four distinct themes: '3C Challenges' addressing communication, collaboration, and coordination complexities with seven specific codes; 'Community Smells' revealing identifiable patterns of community issues through six codes; 'Organizational Challenges' reflecting systemic and structural issues with six codes; and 'Social Confines' encompassing limitations imposed by social dynamics and behaviors within the development environment with five codes.

6. **Producing the report:** In the final phase, we synthesized the thematic analysis findings into a comprehensive report, including the results, discussion, and conclusion sections of the case study. This report detailed the identified themes and their respective codes, explaining their contribution to understanding SD in LSAD. By illustrating the interconnectedness of these themes and their impact on LSAD, our report not only captured the complexity of the issues but also laid a foundation for addressing SD effectively. The synthesis aimed to enhance understanding of SD, its causes, and consequences in LSAD, providing insights to inform mitigation and management strategies in the field.

4 Results

4.1 What are the types of social debt, their causes, and consequences in LSAD?

4.1.1 3C Challenges leading to social debt

Large-scale agile projects, although promising improved flexibility and adaptability, can also face several communication, collaboration, and coordination challenges. These challenges can ultimately lead to the accumulation of social debt. We break down the theme communication, collaboration, and coordination challenges that contribute to social debt in large-scale Agile projects into the following codes :

Hierarchical communication challenges in LSAD stem from top-down channels, clashing with agile values. This leads to unequal team involvement, low morale, and discomfort among diverse employees from different cultures, accumulating social debt. This accumulated social debt causes productivity loss and frustration. One of the interviewees highlighted the hierarchical communication issues, stating, ***" Because you are from different backgrounds and cultures, there could be some hierarchy issues, like people not really speaking up because they expect the person higher up in the hierarchy to speak up... That can be a bit different than occasionally challenging everyone to speak up during meetings because it's not common"[TE11].***

Remote inclusive communication challenges In LSAD, equitable interaction among globally dispersed teams is hindered due to time zone disparities, locations, cultural diversity, and language barriers. These challenges intensify as large-scale agile projects seek global talent to target the worldwide market. Inadequate strategies for inclusive remote communication accumulate social debt, causing disengagement, weakening connections, reduced collaboration, stress, and declining innovation. One of the interviewees effectively expressed the inclusive Communication Challenges in Remote Settings as follows. "***We're often remote or at home, dealing with team members sporadic availability. To handle this, I made a schematic that did not work, which we talked about in person online, which added stress due to scheduling challenges. Building trust online is trickier than face-to-face interactions, impacting our understanding of each other's well-being. Additionally, online discussions can struggle to engage shy team members effectively"[TE5].***

Lack of personal interaction in inter-team communication in LSAD refers to communication between different teams within an organization that lacks a deeper level of engagement, stemming from factors like bad organizational structures, culture differences, and scheduling issues. This led to the formation of social debt. In this context, social debt refers to a sense of owing or being owed communication or assistance to other teams. This

social debt accumulation impedes information flow, coordination, and alignment between teams in the organization. One of the interviewees explained this challenge as follows: "***Different teams are collaborating with us that work on the base-band side since we're testing things. But we don't get enough personal interaction. This lack of personal connection makes it difficult to fully understand the context of the requests sent by different individuals, leading to challenges in effectively providing assistance" [TE2]***.

Hard to communicate people's needs in LSAD refers to the challenge of conveying diverse stakeholder requirements across multiple teams and locations. This challenge arises from agile scaling complexities, evolving requirements, resistance to change, lack of documentation, and deficient requirements elicitation techniques. Unmet user needs can result in social debt, causing extra efforts to address them, leading to increased project cost, scope creep, and stakeholder dissatisfaction. One of the interviewees expressed the challenge as follows: ***"I think it's tough to know how to communicate what people need exactly and what information is required for that. So, there are challenges to effective communication, which can cause social debt because they say, We didn't know about this, and later express that they were unaware of certain matters. We see how unclear communication can result in misunderstandings" [TE1].***

Missing cross-program coordination refers to the lack of a process of bringing together inter- and intra-team members to achieve unity of action in pursuing common goals. One common outcome of missing cross-program coordination is the duplication of efforts. The absence of cross-program coordination within an organization can give rise to the gradual accumulation of social debt because decision-making processes are often delayed when teams do not coordinate effectively. One of the interviewees, who perceives his role as a technical coordinator, discusses the concept of missing cross-program coordination and its implications. ***" There are the kinds of missing things like this cross-program coordination because no one else is doing right... Three trouble reports are in the same area because people are not working together. Which can lead to inefficiencies, duplicated efforts, missed opportunities, and conflicts, ultimately accumulating over time and hindering the overall process's effectiveness [TE7]***.

Lack of coordination for shared vision and active participation refers to the challenges associated with the absence of a well-defined approach to aligning a common vision among team members. It often results in vision misalignment, where different team members interpret the organization's goals differently. That erodes trust and morale among team members, resulting in social debt. One of the interviewees mentioned the difficulty of getting people engaged in meetings and encouraging them to express their opinions. He highlighted a lack of active participation and expressed that some individuals might dominate discussions, leading to uneven engagement levels. ***"I still think one of the challenges is getting people to be more engaged in meetings, speak up, and have their own opinions. And it can be expected that certain people drive the conversation, leading to uneven engagement levels... Coordinating, understanding, and having a collective commitment to the organization's overarching goals is hard"[TE11]***.

Balancing collaboration in remote and hybrid work environments refers to the challenges associated with fostering effective teamwork, communication, and cooperation among employees who work in different locations. Various adverse consequences of remote work include reduced productivity, hindered innovation, decreased team morale, and compromised trust. This gradual loss of meaningful interactions accumulates over time, leading to the accumulation of social debt. One of the interviewees elaborates on the remote work challenges as follows: ***"In remote work, team cooperation suffers because of ineffective collaboration. Having a meeting in person is completely different from having***

it over an hour or so. The dynamics of a face-to-face meeting differ significantly from those held online. You can look at someone when they're working; observing someone's emotions and mood while they work is vital for addressing issues promptly or providing support" [TE2].

4.1.2 Community smells leading to social debt

Due to misalignment, interdependence, and unclear project goals, large agile projects can foster community smells. Community smells that cause social debt in large agile projects are represented in detail in Table 3 as follows:

Lack of psychological safety refers to team members feeling pressured or hesitant to voice their opinions due to fear of criticism. Missed opportunities occur when team members withhold valuable insights and ideas due to fear of criticism, hindering collaboration and optimal outcomes, and resulting in social debt. An interviewee cited a lack of psychological safety and feeling pressured to catch up with the team as contributing factors. ***" I feel immense pressure on our hard working team, leading me to try and match their efforts. This pressure makes new members like me hesitant to seek help due to high expectations. I believe more support and resources are needed for smoother integration into my role and the team" [TE13].***

Black Cloud Effect is when an individual or team's negative attitude harms community morale, causing stress, frustration, and burnout, resulting in social debt. An interviewee highlighted unproductive meetings with unanswered questions and hindered discussions as a manifestation of this effect, causing continuous frustration and wasted time. ***"Meetings, sometimes just 6 minutes long, seem like a mere formality and disrupt other work. They are often perceived as interruptions, contributing to frustration and inefficiency" [TE5].***

Organizational silo effect refers to inefficiencies stemming from fragmented or isolated work practices within the teams or organization. This can result in various negative consequences, such as duplication of efforts and inefficient resource allocation. The cumulative impact of the silo effect leads to the accumulation of the social debt. One of the interviews discusses that issues arise due to people not working together effectively, leading to inefficiencies and conflicts. ***"There are three trouble reports in the same area because the people are not really working together. Which can lead to inefficiencies, duplicated efforts, missed opportunities, and conflicts-ultimately accumulating over time and hindering the overall process's effectiveness[TE7].***

Lone wolf effect refers to individuals who work in isolation to avoid interactions. They tend to isolate themselves from the larger team or community, leading to missed opportunities. This isolation can result in the creation of information silos. This contributes to accumulating social debt by straining relationships and hindering effective teamwork. One of the interviewees discusses his experiences trying to provide information to colleagues who do not actively seek it. ***"When you provide information to someone who doesn't look for information or only mind his own business rather than to get focus on the entire project perspective, who doesn't have questions about the common or shared values and goals, at least from my case, it just goes in one ear, and it goes out the other"[TE8].***

Radio silence effect refers to interactions becoming superficial. This effect can occur when there are only brief updates and deeper discussions are neglected. As a result, important opportunities for decision-making are limited, contributing to the social debt. One of the interviewees mentioned that a scrum master or another person leads meetings primarily for reporting purposes, leading to superficial interactions. ***"Because it will be like a scrum***

Table 3 SD type (Community smells challenges): Causes and effects

Community Smells Challenges			
No	**Causes of SD**	**Effects of SD**	**ID**
1	Lack of Psychological Safety	Lack of psychological safety in teams leads to missed opportunities, hindered collaboration, and members withholding insights.	P13
2	Black Cloud Effect	The Black Cloud Effect leads to stress, frustration, burnout, unproductive meetings, and hindered discussions.	P5
3	Organizational Silo Effect	Organizational Silo Effect leads to duplicated efforts, inefficient resource allocation, and conflicts.	P7
4	Lone Wolf Effect	The Lone Wolf Effect leads to challenges in disseminating information, affecting the project's collective focus and shared goals.	P8
5	Radio Silence Effect	The Radio Silence Effect leads to limited decision-making, superficial interactions, hindered collaboration, reduced team cohesion, and a decline in morale and innovation.	P5
6	The Power Distance Effect	The Power Distance Effect creates a hierarchical dynamic leading to unequal decision-making, role confusion, delayed consensus, and withheld concerns.	P4

master or someone else that is reporting a few things, and it's for teams, sometimes the meeting has been like 6 minutes because you just report, and then it's over. So, it's only like a disturbance with no real discussions" [TE5].

The Power distance effect refers to an organizational phenomenon with a noticeable hierarchy of power and authority. It can lead to unequal decision-making. Unresolved issues can emerge as lower-ranking individuals withhold concerns and accumulate social debt. One of the interviewees explained that higher-level entities or people with authority over these aspects might impact decision-making and resource allocation. ***"We have a line manager, or maybe we have a product manager, and it could be difficult to make decisions, especially when they are a little higher in authority. We have a technical coordinator, but his responsibilities overlap with the existing roles of product owners, leading to confusion about decision-making boundaries. Sometimes, decision-making delays occur as consensus becomes necessary among multiple [higher authority] roles [TE4]***.

4.1.3 Organizational challenges leading to social debt

Organizational challenges cause social debt in LSAD. Transitioning from hierarchical structures to agile methods, cultural shifts, and resistance to new methods produce social debt. We break down the organisational challenges contributing to social debt in LSAD represented in detail in Table 4 as follows:

Organizational fragmentation refers to the condition where an organization is divided into discrete parts that operate independently. In such a state, different departments, teams, or units within the organization may lack cohesion and synergy. This can result in inefficient resource allocation and misaligning goals and objectives, leading to social debt. One of the interviewees explained organizational fragmentation through the term "scattered organization." ***"Our organization is all over the place, very scattered. It's a bit confusing, and I'm unsure how to fix it. One prevalent problem is different departments and teams operate in isolation. Because of this, we face inconsistent procedures across departments. We should also be curious about what's happening in other parts of the organization to stay more Up-to-date"[TE7]***.

Organizational cultural differences refer to the distinct variations in cultural norms within an organization. These differences stem from various regional, national, or demographic factors. Organizational cultural differences can lead to the accumulation of social debt because of conflicts stemming from cultural differences, resistance to change and the formation of cliques or silos based on cultural affinity. One of the interviewees explains how working with people from different cultures can make things a bit tricky at their job. ***"Working with clients and coworkers worldwide can be challenging, especially when everyone has different ideas about how things should be done. For instance, Polish people might seem more direct than Swedes. On the other hand, the Chinese tend to have a more assertive approach. Whether they're managers, Indians, or trainees, their way of working can be quite different. Some cultures have a more hierarchical structure, while we follow a flatter approach here"[TE4].***

Organizational turbulence causing delivery delays refers to an organisation's instability within its operations and decision-making processes. This creates an environment of uncertainty and anxiety among employees as they grapple with the consequences, such as financial setbacks and operational inefficiencies. Combined, these factors foster a negative atmosphere and a buildup of social debt. One of the interviewees highlighted a specific

Table 4 Organization social challenges causes and effects

Organization Social Challenges			
No	**Causes of SD**	**Effects of SD**	**ID**
1	Organizational fragmentation	Organizational fragmentation causes inefficiencies, misalignment of goals, procedural inconsistencies, and insufficient inter-departmental awareness.	P7
2	Organizational cultural differences	Organizational cultural differences lead to conflicts, resistance to change, and the formation of cultural cliques or silos.	P4
3	Organizational Turbulence Causing Delivery Delays	Organizational Turbulence leads to uncertainty and anxiety among employees, financial setbacks, operational inefficiencies, and a negative atmosphere.	P2
4	Trade-offs in Organizational Workflow	Organizational Workflow Trade-offs often cause bypassing ideal processes for urgent objectives, affecting team morale and trust, and sacrificing quality for speed.	P4
5	Organizational Complexity and Management Inefficiencies	Organizational Complexity and Management Inefficiencies lead to unclear decision-making, operational delays, resource misallocation, and increased workloads.	P7

instance of a delivery stop lasting over a month due to what they describe as "bad organizational decisions. ***"We found ourselves in a tough situation where we couldn't deliver our work for over a month because of some bad organisational decisions. So, we have had a delivery stop for over a month. This delay also made it tricky to test our work on real systems in Europe, which added to the challenge"[TE2]. Trade-offs in Organizational Workflow*** refers to an organisation's compromises when faced with conflicting priorities. These trade-offs involve sacrificing certain aspects of an ideal workflow to achieve other objectives. The discrepancy in understanding of the trade-offs creates social debt as individuals question the reliability of the workflow. One of the interviewees described the organization's struggle to adhere to an ideal workflow that perfectly aligns with their requirements. ***"We want to follow this perfectly designed way, but sometimes we can't follow it, so we make trade-offs. If we need to deliver something quickly, getting it to the market quickly is super important. We might have to leave out steps by skipping some important steps"[TE4]. Organizational Complexity and Management Inefficiencies*** refer to the presence of intricate structures within an organization that makes its operations less efficient. It entails a situation where there are excessive layers of management. It leads to social debt by causing unclear decision-making, operational delays, resource misallocation, and increased workloads. One of the interviewees expressed management inefficiencies and difficulties as follows: ***" Our product is complex both in terms of its technical aspects and how we organize things in our organization. When we set our goals, it's difficult to figure out how many components of the project are connected ahead of time. We always miss or inefficiently manage some things because of the project and organization complexity levels"[TE7].***

The organizational challenge of "Bridging Generational Gap" refers to the difficulty of creating a workplace that accommodates the diverse working preferences of employees from different age groups. The tension between flexibility and rigidity in work processes creates conflicts and division among team members, contributing to social debt. One of the interviewees expressed this challenge as follows: ***" If we can work faster, we should try to do so, especially when dealing with older team members or those who come from the 1960s. Getting them to embrace new ways of working can be challenging, as they may not expect to keep working for so long. Additionally, other factors slow down our progress, such as new people like agile to follow, but old employees who like their old ways of working" [TE4]***.

4.1.4 Social confines leading to social debt

Social confines refer to limitations in social interactions, resulting in Social Debt. This term signifies the adverse effects of these restrictions on project performance and outcomes. Social confines challenges causes and effects are represented in detail in Table 5.

Adapting to varied expectations of roles challenge in LSAD occurs when individuals take on conflicting extra roles, creating ambiguity and causing stress and frustration, leading to social debt. An interviewee elaborated on this challenge as follows: ***"The technical coordinator role is quite difficult because it varies across different programs and areas. However, this role is not the same in every part of the organization (Company X). This can be confusing because we don't always know exactly what to do. Sometimes this leads to a lot of confusion in my tasks and feelings of stress and pressure to perform many things in parallel"[TE7].***

Table 5 Social confines challenges causes and effects

Social Confines Challenges			
No	**Causes of SD**	**Effects of SD**	**ID**
1	Adapting to Varied Expectations of Roles	Adapting to varied expectations of roles leads to ambiguity, stress, and frustration.	P7
2	Social Dynamics of Task Complexity	Social Dynamics of Task Complexity can lead to misunderstandings, misaligned expectations, and strained interpersonal relationships.	P2
3	Social Constraints in a Multifaceted Environment	Social Constraints in a Multifaceted Environment lead to unmet social obligations and expectations, unanswered communications, missed meetings, and neglected tasks as individuals struggle to balance their various responsibilities.	P7
4	Lack of Personal Initiative Causing Social Limitations	Lack of Personal Initiative Causing Social Limitations lead to limited personal and career growth, strained relationships, missed learning opportunities, and preventing individuals from realizing their full potential.	P11
5	The Challenge of Integrating the Isolated Team Members	The Challenge of Integrating Isolated Team Members, focusing on effectively engaging socially isolated individuals within a team leads to reduced contributions from these members, harms team cohesion and trust, and lowers overall team performance.	P11

Social dynamics of task complexity in LSAD refers to examining how social interactions and group dynamics influence responses to complex tasks, highlighting the role of misunderstandings, expectations, and interpersonal relationships in creating social debt. An interviewee details these challenges as follows: ***"Some people in our group were curious about why certain tasks took a long time to complete. But then, they were asked to do those complicated tasks by themselves, and that's when they understood why it was taking so much time and was harder to finish quickly. They stopped complaining because they experienced the complexity of these tasks" [TE2].***

Social constraints in a multifaceted environment in LSAD refers to the difficulties of managing a heavy workload, multitasking, and prioritizing tasks while dealing with social dynamics such as emails, meetings, and other professional interactions. Unmet social obligations and expectations, such as unanswered emails, missed meetings, and incomplete tasks, accumulate over time, leading to social debt. One of the interviewees describes the challenge as follows: ***" Imagine having a big email inbox and many meetings. Your inbox keeps filling up even during meetings, and you're trying to keep up. When you concentrate on one thing, you end up neglecting other tasks. This happens naturally in a job where you're involved in many different things and social constraints. Given the various tasks, figuring out what to work on is a daily struggle"[TE7].***

Lack of personal initiative causing social limitations refers to when a person's hesitation to take proactive action in their professional or social life hinders their advancement, opportunities, or relationships. A lack of self-initiative can limit growth, career, and relationships. This apathy wastes learning and collects social debt. One of the interviewees described this challenge as follows. ***"You must be a self-starter and not wait for everything to be handed to you easily... So, the key to succeeding and breaking free from social limitations is to have the drive to take action on your own. The most significant outcome of lacking personal initiative is failing to realize one's full potential" [TE11].***

The challenge of integrating the isolated team members refers to engaging socially isolated team members effectively. Failing to do so can reduce their contributions, harm team cohesion and trust, lower performance, and result in social debt. An interviewee mentioned that introverted and outspoken individuals may struggle with integration. ***"We are a mix of really strong-willed people and a mix of people that maybe like to merge and stay isolated. And then it becomes like we are missing someone in between who can actually get both sides in parallel and keep engaging the people who are not very social or only to themselves. So that's something that we lack. There is the absence of a mediator or facilitator who can bridge the gap between these two groups and work together [TE11].***

4.2 How is social debt linked to TD in the case X context?

In the software development process of Case X, SD and TD are deeply interconnected. The dynamics within the team and the broader organizational culture significantly influence the accumulation of TD. For instance, challenges in communication and coordination within the team lead to a misalignment in understanding project requirements. This misalignment often results in the creation of features that don't quite meet user needs or necessitate substantial rework, directly contributing to technical debt. Similarly, negative team dynamics, such as poor collaboration or working in silos, lead to inconsistencies in coding and problem-solving approaches. These issues not only hinder efficiency but also lead to redundant efforts and suboptimal solutions, further exacerbating the TD. Additionally, resistance to adopting new methodologies or embracing technological changes, often stemming from a

rigid organizational culture or hierarchical structure, slows down progress and innovation. As a result, outdated practices and systems continue to be used, adding to the TD. Under the pressure to meet tight deadlines, teams in Case X might prioritize rapid delivery over long-term quality. This approach often leads to taking shortcuts in the development process, such as insufficient testing or inadequate documentation, culminating in a codebase that is more prone to bugs and difficult to maintain, a clear manifestation of increasing TD. Moreover, high employee stress levels and turnover rates result in a loss of institutional knowledge and continuity, causing errors and inconsistencies in software development, further contributing to TD. The lack of investment in ongoing training and skill development leaves the team ill-prepared to handle current and future technological challenges. This gap leads to reliance on outdated or inefficient practices, which in turn contributes to the accumulation of TD. Below is a comprehensive breakdown of how each identified type of SD is linked to TD in the context of Case X.

4.2.1 3C Debt leading to TD

In Case X, the 3C challenges in LSAD distinctly contribute to TD through various cause-and-effect mechanisms. Hierarchical communication challenges conflict with agile values, decreasing productivity and creating decision-making bottlenecks, resulting in delayed software releases and the need for later modifications to address suboptimal features, contributing to TD. Remote communication challenges hinder effective interaction, leading to the development of suboptimal solutions, necessitating rework and adjustments, and contributing to TD through code refactoring and design changes. Inter-team communication barriers impede information flow and coordination, resulting in misalignments in understanding project requirements and goals, leading to inconsistencies in code and architecture, manifesting as integration efforts and additional testing requirements, contributing to TD. Stakeholder communication issues, due to agile scaling complexities and evolving requirements, result in unmet user needs and project scope creep, leading to scope changes, budget overruns, and TD. Cross-program coordination deficits cause duplication of efforts, redundant code, and inefficient utilization of team efforts, adding to the TD. Inadequate coordination for a unified vision and active participation among team members leads to vision misalignment and inconsistent project progress, resulting in delays and fragmented software functionality, further contributing to TD. Challenges in fostering effective teamwork in remote and hybrid work environments reduce productivity, leading to misunderstandings in requirements and inefficient development practices, exacerbating TD.

4.2.2 Community smells debt leading to TD

In Case X, various community smells contribute to TD in software development. The lack of Psychological safety hampers open communication, stifling innovation and optimal decision-making, leading to suboptimal features and rework. The Black Cloud Effect fosters negative team attitudes, leading to unproductive meetings, stress, and delays. The Organizational Silo Effect causes duplication of efforts and inefficient resource allocation, resulting in redundant code and misaligned features. The Lone Wolf Effect, with individuals working in isolation, creates information silos and integration challenges. The Radio silence effect limits meaningful communication, leading to misaligned project objectives and overlooked improvement opportunities. Lastly, the Power Distance Effect delays decision-making, resulting in suboptimal choices and technical inefficiencies. Each

community's smell impedes communication, collaboration, and decision-making, exacerbating TD in software development.

4.2.3 Organizational social debt leading to TD

Organizational challenges significantly contribute to SD issues, leading to TD. Organizational fragmentation, dividing the organization into isolated units, causes inefficient resource allocation and misaligned goals, requiring harmonization efforts and increased integration, resulting in TD due to different workflows and system incompatibilities. Cultural differences within the organization create internal conflicts and resistance to change, leading to cliques or silos and TD due to a lack of cohesive strategy in software development. Diverse methodologies and practices across the organization necessitate alignment efforts, increasing complexity and workload in the software development process. Organizational turbulence, characterized by unstable operations and decision-making, causes uncertainty and delivery delays, leading to TD through project hold-ups and rushed development efforts, compromising software quality and adherence to best practices. Trade-offs in organizational workflow, driven by conflicting priorities, often involve skipping essential processes like testing, leading to TD by increasing defects and future rework needs, compromising software reliability. Complex organizational structures and management inefficiencies result in unclear decision-making and operational delays, causing TD through resource misallocation and increased workloads, leading to inefficiencies in the software development lifecycle. Bridging the generational gap and accommodating diverse working preferences of different age groups contributes to TD by slowing the adoption of efficient work processes, leading to reliance on outdated methods and hindering innovation and competitiveness in software solutions.

4.2.4 Social confines debt leading to TD

Ambiguity in roles and expectations creates stress and confusion, leading to TD through project delays and inefficiencies. Misaligned objectives and duplicated efforts slow down development and increase error risks. Social dynamics affect handling complex tasks, causing stress and inefficiencies. This contributes to TD by causing delays and quality issues due to ineffective coordination. Social constraints in a multifaceted environment lead to unmet obligations and neglected tasks, causing TD, as critical tasks are overlooked, creating bottlenecks and increasing error risks. Lack of personal initiative hinders learning and team contributions, delaying project progress and reducing software quality, leading to TD. Integrating isolated team members poorly reduces contributions and harms cohesion, creating information silos and inconsistencies, and affecting project efficiency and consistency.

4.3 How to mitigate SD in LSAD?

3C Challenges: mitigation strategies: Dreesen et al. (2021) proposed four preventive actions for managing communication, collaboration, and coordination challenges. They emphasized honest and open communication, reduced upfront interaction costs, and improved psychological safety. We provide mitigation strategies for 3C Challenges subtypes in Case X (see Table 6). Researchers Kolesnikova (2023) suggested fostering open communication to enhance information flow across hierarchical levels. In our analysis of the literature, we found that clear and concise language is crucial for accurate

Table 6 3C challenges: Mitigation strategies

Theme (SD-type): 3C Challenges: Mitigation Strategies			
No	**Subthemes**	**Strategies**	**Source**
1	Hierarchical communication challenges	Create a culture of open communication	Kolesnikova (2023)
		Use clear, concise language when communicating	
		Set up regular feedback mechanisms	
2	Remote inclusive communication challenges	Effective use of communication tools	Walimbe (2016)
		Effective daily scrum meetings	
3	Lack of personal interaction in inter-team communication	Use simple language	Gratis (2022)
		Practice active listening	
		Stay calm	
		Provide feedback	
4	Hard to communicate	Workshops and focus groups with end users	Kasauli et al. (2021)
		Feedback and clarification	
		Building long-lasting customer knowledge	
5	Lack of coordination for shared vision and active participation	Dependency awareness across development teams	Bick et al. (2017)
		Goal-setting framework for shared vision and goals (OKRS)	
6	Balancing collaboration in remote and hybrid work environments	Create smaller subteams within existing teams	Reunamäki and Fey (2023)
		Promote engagement in digitally mediated agile meetings	
		Protect non-meeting time within the organisation	
		Be truly present and build opportunities to interact with employees	

understanding, reducing misinterpretation and delays. Feedback mechanisms like retrospectives facilitate continuous improvement and address communication gaps. Encouraging bottom-up communication empowers team members, fostering inclusive decision-making. Training in effective communication skills promotes clear expression of ideas and active listening, fostering collaboration. To manage remote teams, using communication tools effectively and ensuring seamless information sharing is also recommended by researchers Walimbe (2016). In literature, various strategies are proposed to address communication and coordination challenges in team environments. Daily scrum meetings, with flexible scheduling to accommodate time zone differences, maintain alignment and momentum Gratis (2022). Overcoming language barriers through simplified language and translation aids enhances inclusivity. Promoting coordination through defined roles and collaborative tools ensures synchronization Bick et al. (2017). Fostering trust through transparency and consistent communication encourages open dialogue and problem-solving. Using simple language for clear understanding is suggested for personal interactions in inter-team communication. Active listening, staying calm during discussions, and providing constructive feedback are crucial for maintaining a positive communication climate. Holding frequent and transparent meetings with the whole team promotes unity and ensures alignment on goals and progress Kasauli et al. (2021). To address user needs, conducting workshops and focus groups with end users is recommended, along with regular feedback and building a repository of customer knowledge Kasauli et al. (2021). Fostering dependency awareness across development teams and implementing goal-setting frameworks align teams towards shared visions and goals Bick et al. (2017). Balancing collaboration in remote and hybrid work environments involves creating smaller subteams, promoting engagement in digital meetings, protecting non-meeting time, fostering employee interaction, adopting software solutions, and optimizing task allocation between home and on-site work Reunamäki and Fey (2023). Additionally, creating areas in the workplace that encourage informal interactions and team bonding strengthens team dynamics.

Community smells mitigation strategies: We are summarising the mitigation strategies for community smells (see Table 7). In community smells; one notable challenge is the lack of psychological safety. To address this, the researchers Newman et al. (2017) suggest several strategies: providing support from leadership, setting clear expectations for roles and responsibilities, fostering a culture valuing constructive feedback, offering training on emotional intelligence and conflict resolution, appreciating team members' contributions, and conducting surveys to gauge psychological safety levels. Another significant challenge is the Black Cloud Effect. To counter this, the research recommends Lambiase et al. (2022) identifying root causes of negativity, shifting focus towards solutions, and proactively addressing underlying issues. These strategies can help create a positive and productive work environment conducive to collaboration and innovation. Further organizational silos present a challenge, hindering communication and collaboration between departments or teams. Strategies to mitigate this include Mouta and Meneses (2021) aligning departmental goals with organizational objectives, empowering middle managers to facilitate communication, and securing senior leadership commitment. The Lone Wolf Effect arises when some team members prefer working alone, potentially disrupting team cohesion. Understanding the reasons behind this preference, fostering a positive team culture, and celebrating team efforts can address this challenge Catolino et al. (2020). Communication breakdowns, known as the Radio Silence Effect, can impede progress. Strategies to mitigate this include Catolino et al. (2020) establishing clear communication protocols, scheduling regular check-in meetings, and ensuring consistent channels. The Power Distance Effect highlights hierarchical barriers to

Table 7 Community smells challenges: Mitigation strategies

Theme (SD-type): Community Smells Challenges: Mitigation Strategies			
No	**Subthemes**	**Strategies**	**Source**
1	Lack of Psychological Safety	Leadership must provide support.	Newman et al. (2017)
		Clear expectations for roles and responsibilities.	
		Foster a culture of constructive feedback.	
		Training on emotional intelligence and conflict resolution.	
2	Black Cloud Effect	Take the time to identify the underlying causes of negativity within the team or organisation.	Lambiase et al. (2022)
3	Organizational Silo	Align departmental goals and objectives with overall organisational goals.	Mouta and Meneses (2021)
4	Lone Wolf Effect	Take the time to understand the reasons behind any individual team member's preference for working alone.	Catolino et al. (2020)
		Foster a positive team culture where collaboration is celebrated.	
5	Radio Silence Effect	Define clear communication protocols outlining how information should be shared.	Catolino et al. (2020)
6	The Power Distance Effect	Open communication channels, regardless of hierarchical positions.	Saeeda et al. (2023)
		Organize team-building activities.	

communication. Strategies to address this include Saeeda et al. (2023) promoting open communication channels, organizing team-building activities, and creating mechanisms for employee feedback on management decisions. These strategies aim to foster collaboration and drive success in large-scale agile projects.

Organizational social debt: mitigation strategies: Current literature analysis presents a comprehensive set of mitigation strategies aimed at addressing various social challenges within organisations(see Table 8). To counter organisational fragmentation, researchers Annosi et al. (2020) recommend the implementation of a centralised project management system or software coupled with the formation of cross-functional teams. This approach allows for streamlined tracking of tasks, deadlines, and progress while promoting collaboration across different departments or disciplines. In tackling organisational cultural differences, the findings Waisfisz (2015) advocate for targeted training programs focused on understanding diverse cultural norms and communication styles. Furthermore, emphasising common goals and values that transcend cultural disparities can foster a more inclusive and cohesive work environment. To address delivery delays stemming from organisational turbulence, the studies Purna Sudhakar et al. (2011) highlight the importance of robust risk management processes and effective change management strategies. Additionally, ensuring adequate resource allocation and implementing performance monitoring mechanisms are crucial for mitigating potential bottlenecks. In dealing with trade-offs in organisational workflow, the literature analysis Xiao and Peng (2019) underscores the significance of defining project priorities, strategically allocating resources, and providing strong leadership support. Finally, to streamline organisational complexity and inefficiencies, the researchers Diagnostics of the efficiency of the organization management system (2022) suggest simplifying hierarchies, defining clear roles and responsibilities, and empowering team members with decision-making authority.

Social confines debt: mitigation strategies We are providing mitigation strategies for Social confining challenges from the literature(see Table 9). Social constraints within Agile teams present multifaceted challenges that necessitate careful navigation for effective project execution. One significant hurdle is the adaptation to varied role expectations, exacerbated by the dynamic nature of Agile projects where team members often juggle multiple responsibilities. To mitigate this, scholars advocate for clear delineation of roles and responsibilities alongside promoting cross-training and skill development to foster adaptability Burga et al. (2022). Task complexity adds another layer of social constraints, impacting communication, coordination, and decision-making processes within Agile teams. Breaking down complex tasks into smaller, manageable components is recommended to alleviate overwhelm and enhance focus Masood et al. (2020). Leveraging individual strengths and expertise ensures efficient task allocation, optimizing team performance and facilitating project progress. Moreover, fostering inclusivity and embracing diversity within Agile teams are crucial for overcoming social challenges stemming from multifaceted environments. Building trust, valuing diverse perspectives, and celebrating contributions from all team members foster a collaborative culture that enhances creativity and problem-solving Davila et al. (2020). Encouraging personal initiative and actively integrating isolated team members further strengthens team cohesion, ultimately enabling Agile teams to navigate social constraints successfully and achieve project objectives effectively Rahy and Bass (2021).

Table 8 Organizational social challenges: Mitigation strategies

Theme (SD-type): Organization Social Challenges: Mitigation Strategies			
No	Subthemes	**Strategies**	**Source**
1	Organizational fragmentation	Implement a centralized project management system or software that allows for easy tracking of tasks, deadlines, and progress.	Annosi et al. (2020)
2	Organisational cultural differences	Training on focusing on understanding different cultural norms,communication styles, and working preferences.	Waisfisz (2015)
		Identify and communicate common goals and values that transcend cultural differences.	
3	Organisational turbulence causing delivery delays	Implement a robust risk management process to identify potential sources of organizational turbulence early on.	Purna Sudhakar et al. (2011)
		Establish effective change management processes to manage organizational changes that may impact project delivery.	
4	Trade-offs in organisational workflow	Clearly define project priorities and critical path activities. Identify tasks that are essential for project success and focus resources on completing them first. Allocate resources strategically based on project priorities and requirements	Xiao and Peng (2019)
		Identify tasks that are essential for project success and focus resources on completing them first.	
		Allocate resources strategically based on project priorities and requirements	
5	Organisational complexity and management inefficiencies	Streamline the organizational structure by reducing unnecessary layers of hierarchy and bureaucracy.	Diagnostics of the efficiency of the organization management system (2022)
		Define clear roles and responsibilities for team members involved in the project.	
		Empower team members by delegating decision-making authority and autonomy.	

Table 9 Social confines challenges: Mitigation strategies

Theme (SD-type): Social Confines Challenges: Mitigation Strategies			
No	**Subthemes**	**Strategies**	**Source**
1	Adapting to varied expectations of roles	Clearly define the roles and responsibilities within the Agile teams.	Burga et al. (2022)
		This includes understanding the specific tasks and making decisions.	
2	Social dynamics of task complexity	Break down complex tasks into smaller, more manageable chunks.	Masood et al. (2020)
		This makes the tasks more approachable and allows team members to focus on specific aspects without feeling overwhelmed.	
3	Social constraints in multifaceted environment	Foster an inclusive environment where diverse perspectives, backgrounds, and experiences are valued and respected.	Davila et al. (2020)
		Invest time and effort in building trust among team members.	
4	Lack of personal initiative causing social limitations	Communicate expectations regarding personal initiative and accountability within the team or organisation.	Burga et al. (2022)
5	Challenge of integrating isolated team members	Schedule regular check-ins with isolated team members to provide updates, gather feedback, and address any concerns they may have.	Rahy and Bass (2021)

5 Lessons learned: LSAD social debt effects on telecom software business support

Case X's software business faces challenges in adapting to the complexity and dynamic nature of telecom networks. This requires flexibility, rapid problem-solving, and iterative development. Case X's operational processes involve agile methodologies. They also include strategic collaborations with industry organizations. They adopted agile due to its adaptability to the fast-paced telecom industry, the customer-centric approach of close cooperation with network operators, and its efficiency for rapid solution deployment. Agile's emphasis on quality assurance, cross-functional collaboration, transparency, risk mitigation, and continuous improvement enhances Company X's ability to deliver high-quality, client-centric solutions in a dynamic telecom sector. Company X derives numerous benefits from agile adoption but faces scalability limitations related to people, management, team dynamics, and working as projects and organizations grow.

In Case X, organizational challenges in agile adoption necessitate cultural shifts, with employee and stakeholder resistance potentially hindering success. Realignment of structures and processes to accommodate agile principles can face resistance. Social and psychological safety concerns involve managing demanding client expectations and balancing speed and quality, which may strain team performance and create stress during testing and quality assurance efforts in the telecommunications industry. These Social challenges in Company X contribute to the accumulation of social debt, impacting global collaboration infrastructure, organizational practices, and project management approaches in their software business.

Social debts like hierarchical management issues and inter-team communication gaps hinder the deployment and management of end-to-end services, causing delays and inefficiencies. These challenges can affect customer satisfaction and the goal of providing high-quality services. In the Self-Organizing Networks (SON) context, poor cross-program coordination and hierarchical communication can delay self-optimization processes, impacting efficient network operations. Moreover, difficulties in requirement elicitation and sharing a holistic vision can hinder the collection and analysis of performance data for big data analytics, potentially leading to suboptimal machine-learning models and anomaly detection. Additionally, the absence of personal interaction and cross-program coordination among teams can delay the implementation of network-slicing solutions, affecting resource allocation and the ability to offer tailored services to customers.

Community Smells like Lack of Psychological Safety, Black Cloud Effect, Organizational Silo Effect, Lone Wolf Effect, Radio Silence Effect, and Power Distance Effect have diverse adverse effects, notably within LSAD for telecom solutions. These issues can inhibit innovation by discouraging team members from sharing valuable insights and ideas, potentially impeding the adoption of crucial cutting-edge technologies in the telecom sector. The Power Distance Effect in Company X can hinder rapid decision-making in the evolving telecom market, while the Lone Wolf Effect disrupts cross-functional team collaboration. Organizational silo effects lead to inefficiencies and hinder market responsiveness. Missed insights into market demands due to a lack of psychological safety and the Black Cloud Effect can lead to customer dissatisfaction. Power distance and organizational silo effects can compromise efficient network sharing among operators, affecting service quality. The Lone Wolf Effect can cause information silos that make it harder to design user experiences for telecom solutions. Problems with allocating resources can also cause infrastructure upgrades that are needed for telecom projects to be delayed, which can make it harder to meet customer needs and deploy solutions on time.

Organizational complexities and management inefficiencies can hinder the implementation of Next Generation Mobile Networks (NGMN) design principles, potentially causing delays in achieving streamlined operational and management processes. Architectural solutions like Network Functions Virtualization (NFV), Software Defined Networks (SDN), and Mobile Edge Computing (MEC) rely on efficient organization and management, and organizational challenges may lead to inconsistencies, compromises in technology effectiveness, and deployment delays. These issues can impact network scalability, adaptability, traffic management, and the success of technologies like Cloud Radio Access Networks (C-RAN), Internet of Things (IoT) infrastructures, and network slicing in meeting telecom business requirements. Network sharing models like passive sharing, MVNOs, and cloud infrastructure sharing depend on efficient coordination and resource allocation. Still, organizational challenges may hinder their effectiveness, potentially resulting in underutilized resources and higher costs. Social constraints disrupt technology adoption, hinder coordination and collaboration within the organization, and affect team structures, potentially delaying product and service delivery in the market. For telecom business support and excellence, it is vital to identify and mitigate the causes of social debt more efficiently.

6 Limitation and threat to validity

The validity threats were considered with the lens developed by Runeson and Höst (2009). To minimize reliability threats, we involved multiple researchers in the design and execution of this study. Further, the interviewees verified our findings, which helped to avoid incorrect interpretations. Ensuring construct validity involves accurately measuring the concept under study and mitigating misinterpretations of interview questions through clarification and pre-testing. Careful selection of experienced interviewees knowledgeable in agile methods and large-scale projects addresses concerns about sample suitability. However, internal validity remains challenging in software engineering, given the intricate variables influencing project outcomes. Our study identifies multiple causes of social debt impacting LSAD, while key internal validity threats include researcher and interviewee biases. In mitigating these threats, multiple researchers designed interviews and engaged interviewees in various roles. However, external validity may be limited due to the nature of the single case study, restricting generalizability. To address this, we provide a detailed context of our study to minimize this limitation. The study was conducted in Sweden, suggesting potential generalizability to other Nordic countries, but cultural differences may warrant further investigation for validation. Best practices may be applicable over time, but caution is needed due to the risk of researcher interpretation bias. Data collection involved participants with diverse roles and included multiple researchers at each stage to mitigate bias.

7 Discussion and conclusion

Our research highlights the broader impact of social factors in software development, exploring their causes and practical implications, especially within the context of LSAD in secure telecom business support. In LSAD projects, the quest for greater flexibility and adaptability poses multifaceted challenges. If left unaddressed, these challenges accumulate, resulting in social debt, which manifests as community smells, organizational challenges, and social confinement issues Saeeda et al. (2023). We found that community smells and organizational challenges are the leading causes of social debt in LSAD.

Organizational issues like fragmentation, cultural differences, turbulence, workflow trade-offs, complexity, and generational gaps hinder telecom software businesses, impacting resource allocation, collaboration, timelines, quality, decision-making, and productivity. These challenges accumulate as social debt, leading to organizational social debt with complex issues such as structural challenges, integration difficulties, system complexity, and challenging operations Tamburri (2019); Saeeda et al. (2023). Community smells have significant implications for telecom software production. Firstly, a lack of psychological safety in development teams can stifle innovative ideas and process improvements, potentially leading to a lack of product innovation. Secondly, negative team attitudes, such as the Black Cloud Effect, can result in stress, burnout, and reduced motivation among developers, impacting software quality and increasing the likelihood of coding errors and defects, which accumulate as technical debt Tamburri (2019). Community smells like the Lone Wolf Effect and Radio Silence Effect can create information silos within development teams, leading to misunderstandings, missed requirements, and coordination challenges that cause delays and affect software quality. Organizational silos resulting from these community smells can lead to duplicated efforts and conflicts, slowing down development and increasing costs and accumulating process debt Saeeda et al. (2023). The Power Distance Effect, characterized by hierarchical imbalances, can impede decision-making, causing delays and suboptimal choices, further impacting production efficiency. Our study identifies five social confines challenges that impact telecom software production, including role ambiguity and stress, social dynamics affecting team cohesion, multitasking, unmet obligations causing delays, a lack of personal initiative, and the need to integrate socially isolated team members for cohesion. The social debt caused by unmet obligations, ambiguity, and stress further leads to people's debt Saeeda et al. (2023). The social debt arising from isolated team members and a lack of personal initiative also lead to an unparticipatory culture, representing a form of cultural debt Saeeda et al. (2023).

Our comparative analysis between our case study and existing literature Ahmad and Gustavsson (2022); Saeeda et al. (2023); Kazman (2019); Tamburri et al. (2015); Tamburri and Di Nitto (2015); Tamburri (2019) reveals both similarities and differences. Existing literature mainly focuses on defining social debt and its origins in software engineering, particularly within the context of 3C challenges. However, there's a notable gap in the literature exploring social debt within LSAD. The available literature lacks comprehensive coverage of the causes and consequences of social debt in LSAD, with limited empirical validation from industrial studies. Our study confirms known community smells, including the Black Cloud Effect, Organizational Silo Effect, Lone Wolf Effect, Radio Silence Effect, and The Power Distance Effect, in line with existing research. We explored a new community smell lack of "psychological safety" that had not previously been reported. However, we do not find hyper community smell or other smells like priggish members, primadonnas, sharing villainy, and solution defiance, as mentioned in the latest state-of-the-art literature Ahmad and Gustavsson (2022); Saeeda et al. (2023). We identify new organizational challenges and social confinements leading to social debt that were previously unreported. In contrast, Saeeda et al. (2023) discusses different organizational challenges (e.g., development community uncooperativeness, architectural changes) and social confinement challenges (e.g., social isolation, team pressure) not found in our case study.

We explore social debt in LSAD for telecom software support, paving the way for further research. For practitioners, recognizing social debt's impact on LSAD is crucial, encompassing social interactions, team dynamics, and organizational factors. Addressing specific causes like community smells, organizational complexities, and social limitations allows tailored strategies to mitigate social debt. Recognizing community smells helps create inclusive and collaborative work environments. Effective strategies for managing organizational challenges can lead to smoother transitions and reduced social debt accumulation.

8 Future work

This study suggests several future research directions. Firstly, further empirical research is recommended to validate and expand on identified causes and effects of SD in LSAD, enhancing understanding of this phenomenon. Secondly, investigating the influence of leadership changes within organizations on social debt could provide insights into managing smooth transitions. Additionally, exploring how client interactions, especially with challenging clients, contribute to SD in LSAD could be beneficial. Moreover, examining the impact of external factors like market volatility, regulatory changes, and economic downturns on social debt in LSAD would be valuable for understanding its dynamics. In the future, it will also be interesting to investigate the link between the accumulation of unresolved usability engineering issues and social debt, which may negatively impact user satisfaction and efficiency in the long term Saeeda et al. (2018). Comparing findings with other large-scale agile projects or industries could help determine generalizability and identify unique factors. Lastly, researching mitigation strategies for identified social debt causes and their impact on project outcomes is recommended for future work.

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Data availability The qualitative data analysis is provided within the manuscript. The reported data, its analysis, and findings are anonymized according to the Swedish laws supplementing the General Data Protection Regulation (Regulation (EU) 2016/679) (GDPR).

Declarations

Competing interest The authors declare no competing interests.

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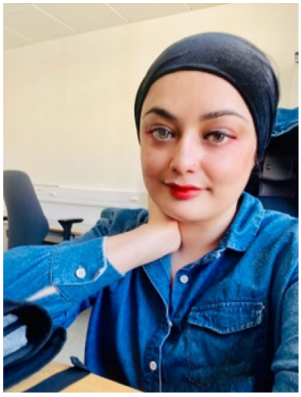

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